Satwik Paladugula

Entry-Level Software Engineer • Java, SQL & AI Systems

Tadepalligudem, AP, India • +91-7674864689 • satwik000001@gmail.com • GitHub • LinkedIn

PROFILE

Computer Science undergraduate with strong foundations in **Java, SQL, Object-Oriented Programming (OOP)**, and **Database Management Systems**. Experienced in building robust backend applications using **Spring Boot, FastAPI, and MySQL**, with an active focus on integrating AI-driven capabilities and secure API services. Passionate about writing clean, maintainable code, collaborative problem solving, and continuous technical learning.

TECHNICAL SKILLS

- **Programming Languages:** Java, SQL, Python, C
- **Backend & Frameworks:** Spring Boot, Spring Data JPA, Hibernate, FastAPI, RESTful APIs
- **Databases & Storage:** MySQL, SQLite, Relational Database Design, Query Optimization
- **Developer Tools:** Git, GitHub, Postman, VS Code, Linux
- **Core CS Foundations:** Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

FEATURED PROJECTS

AI Security Gateway — Enterprise LLM Proxy & Prompt Defense *Python, FastAPI, SQLite, Pydantic, Streamlit*

- Built a high-performance reverse proxy between clients and LLMs to prevent confidential data leakage and defend against prompt injection attacks.
- Implemented algorithmic PII detection (Luhn checksum, Shannon entropy) adding only **0.14 ms p50 latency**, with multi-turn token masking and **90% unit test coverage**.

Student Management System — Enterprise RESTful Backend *Java, Spring Boot, Spring Data JPA, Hibernate, MySQL*

- Designed and implemented a production-grade RESTful API backend using Spring Boot and MySQL for streamlined student record and course management.
- Built full CRUD workflows using Spring Data JPA and Hibernate for reliable transactional data persistence, thoroughly verified via Postman test collections.

ClarifySQL — AI-Powered Natural Language Data Assistant *Python, FastAPI, Gemini API, MySQL, SQLGlot*

- Developed an intelligent data assistant converting natural language business questions into secure, schema-aware SQL queries for MySQL databases.
- Integrated strict AST-level SQL validation using SQLGlot to prevent unauthorized mutations and built a responsive FastAPI service with conversation context.

EDUCATION

B.Tech — Computer Science and Technology **2023 – 2027**
Sri Vasavi Engineering College, Tadepalligudem **CGPA: 8.25 / 10**
Intermediate (PCM) — Bharatiya Vidya Bhavans **2021 – 2023 | 84.8%**
SSC — Tirumala Proactive School **2020 – 2021 | 10 / 10 CGPA**

CERTIFICATIONS

- **ServiceNow Certified Application Developer (CAD)** & **System Administrator (CSA)**
- **SAP Certified Associate — Back-End Developer - ABAP Cloud**
- **Oracle Certified Generative AI Professional** & **Salesforce Agentforce Specialist**